

Fixed point parking

1. Course Content

Note: This program runs on a sandbox map; you will need to adjust the program for your personal map.

1. Learn the actions the car takes after recognizing a stop.
2. Learn the newly introduced key source code.

2. Parking Detection

Road Sign Detection Source Code Path:

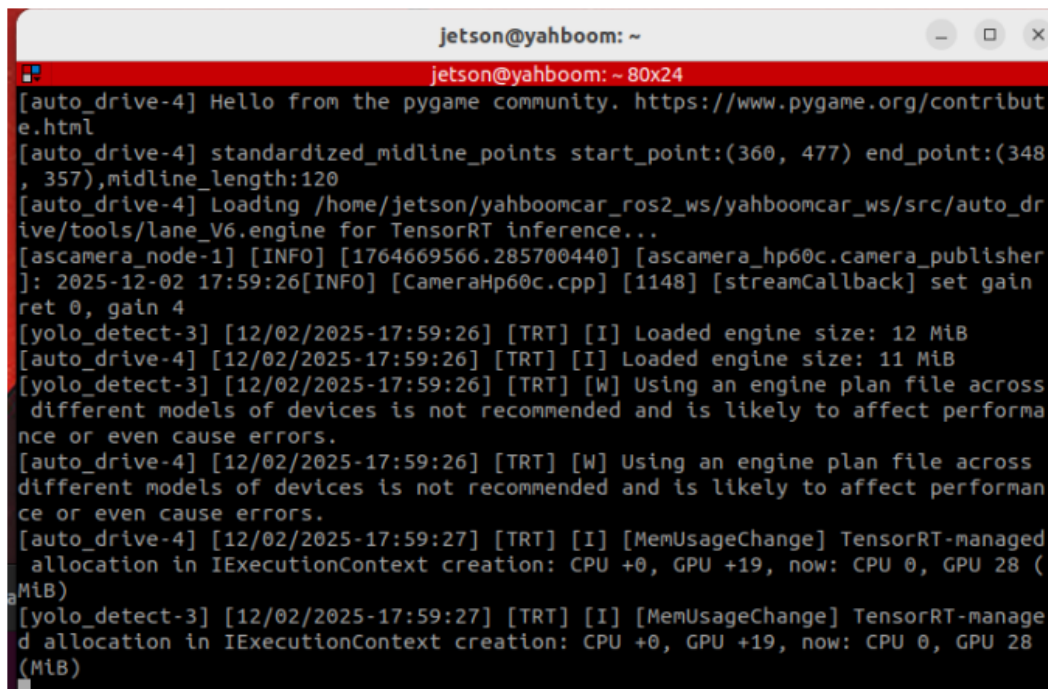
```
/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/yolo_detect.py  
/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/auto_drive.py
```

3. Program Startup

3.1 Startup Command

- Start camera + chassis + road signs, lane lines

```
ros2 launch auto_drive auto_drive_bringup.launch.py
```

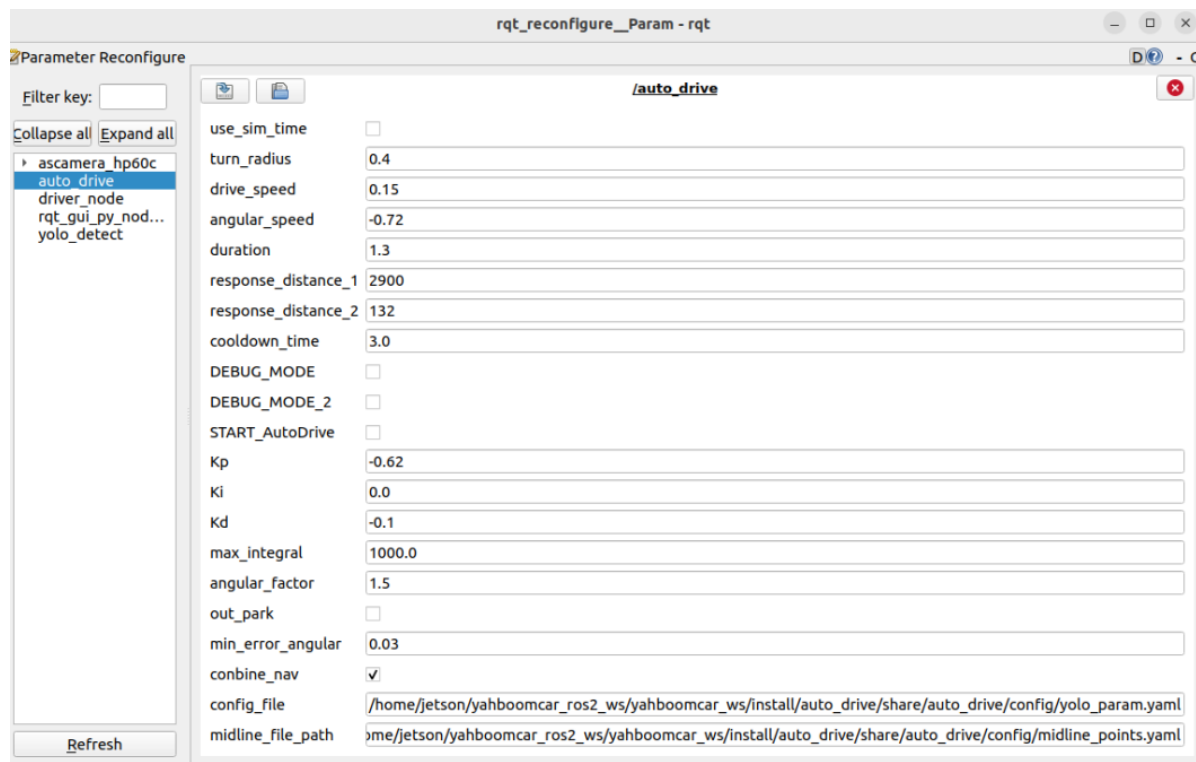
A terminal window titled 'jetson@yahboom: ~' with a red header bar. The terminal shows the output of the command 'ros2 launch auto_drive auto_drive_bringup.launch.py'. The output includes messages from 'auto_drive-4', 'ascamera_node-1', and 'yolo_detect-3'. Key messages include: 'Hello from the pygame community', 'standardized_midline_points start_point:(360, 477) end_point:(348, 357), midline_length:120', 'Loading /home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/tools/lane_V6.engine for TensorRT inference...', 'Loaded engine size: 12 MiB' and '11 MiB', and 'Using an engine plan file across different models of devices is not recommended and is likely to affect performance or even cause errors.' The terminal also shows memory usage changes for TensorRT-managed allocation in IExecutionContext creation, indicating CPU and GPU memory usage.

After the program runs, it will display the YOLO recognition screen. The car will run in the center of the lane line and perform corresponding actions based on the road signs it recognizes during the journey. (If the terminal doesn't report any errors after starting the program, but the robot doesn't move, you can run the following program to check the parameters.)



- Start the dynamic parameter configuration node

```
ros2 run rqt_reconfigure rqt_reconfigure
```

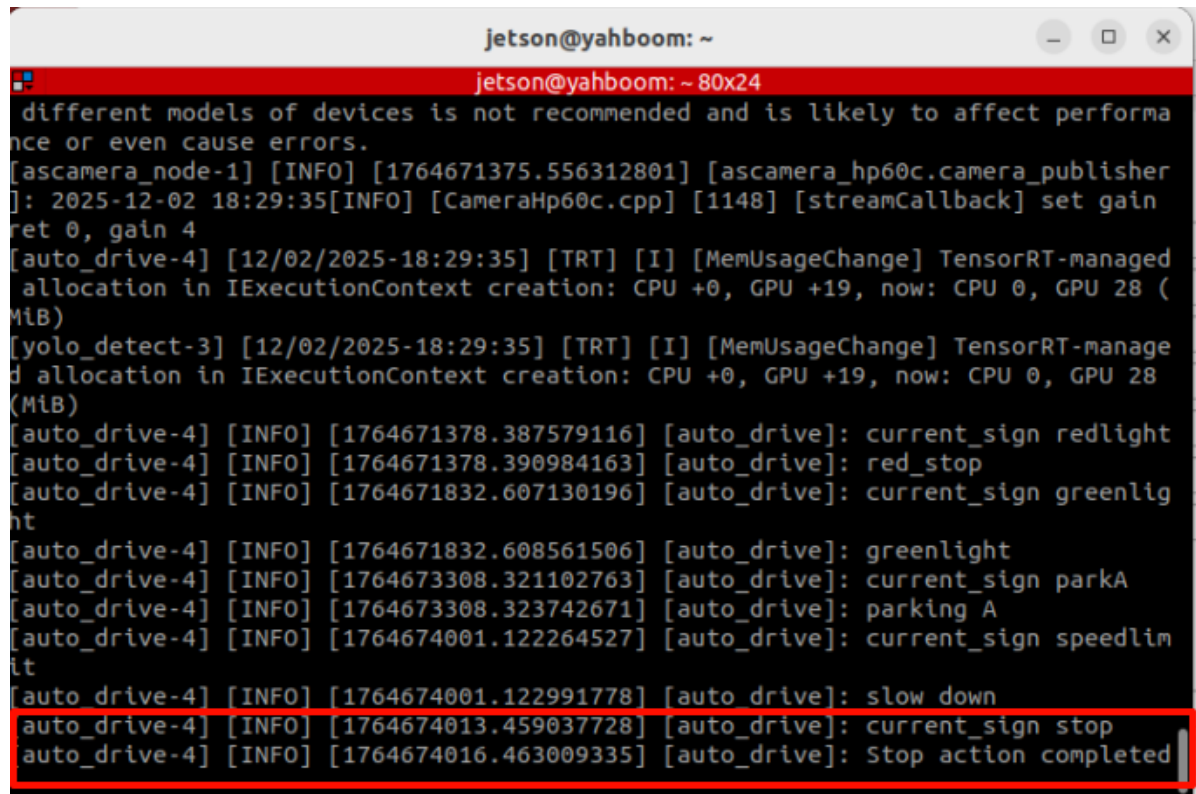


If the robot doesn't move, check two parameters: Is START_AutoDrive checked? Is combine_nav false?

✓ After modifying the parameters, click on the blank space in the GUI to enter the parameter values. Note that this will only take effect for the current startup; for permanent effects, you need to modify the parameters in the source code.

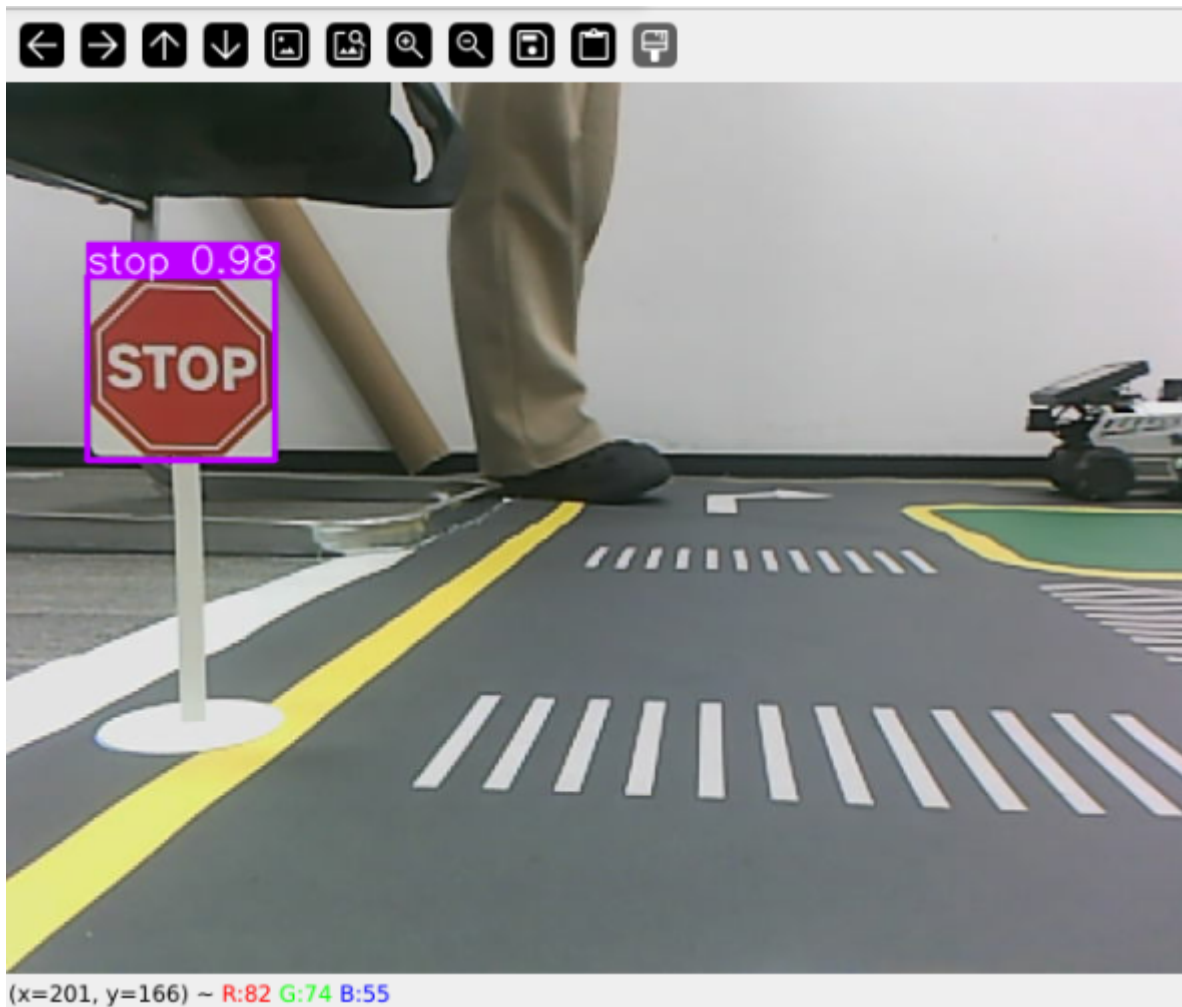
3.2 Car Movement After Parking Detection

The program will print "stop" when it detects parking.



```
jetson@yahboom: ~
jetson@yahboom: ~ 80x24
different models of devices is not recommended and is likely to affect performance or even cause errors.
[ascamera_node-1] [INFO] [1764671375.556312801] [ascamera_hp60c.camera_publisher]: 2025-12-02 18:29:35[INFO] [CameraHp60c.cpp] [1148] [streamCallback] set gain ret 0, gain 4
[auto_drive-4] [12/02/2025-18:29:35] [TRT] [I] [MemUsageChange] TensorRT-managed allocation in IExecutionContext creation: CPU +0, GPU +19, now: CPU 0, GPU 28 (MiB)
[yolo_detect-3] [12/02/2025-18:29:35] [TRT] [I] [MemUsageChange] TensorRT-managed allocation in IExecutionContext creation: CPU +0, GPU +19, now: CPU 0, GPU 28 (MiB)
[auto_drive-4] [INFO] [1764671378.387579116] [auto_drive]: current_sign redlight
[auto_drive-4] [INFO] [1764671378.390984163] [auto_drive]: red_stop
[auto_drive-4] [INFO] [1764671832.607130196] [auto_drive]: current_sign greenlight
[auto_drive-4] [INFO] [1764671832.608561506] [auto_drive]: greenlight
[auto_drive-4] [INFO] [1764673308.321102763] [auto_drive]: current_sign parkA
[auto_drive-4] [INFO] [1764673308.323742671] [auto_drive]: parking A
[auto_drive-4] [INFO] [1764674001.122264527] [auto_drive]: current_sign speedlimit
[auto_drive-4] [INFO] [1764674001.122991778] [auto_drive]: slow down
[auto_drive-4] [INFO] [1764674013.459037728] [auto_drive]: current_sign stop
[auto_drive-4] [INFO] [1764674016.463009335] [auto_drive]: Stop action completed
```

The car will stop at a fixed point for three seconds, then resume driving along the line.



The fixed-point parking handling function is located in the `auto_drive.py` file. Below is the function entered after parking is detected. Users can modify it to reflect the car's movement after detecting road signs.

```
def stop(self):
    '''停止 stop'''
    self.turn_action_active = True # Set an action sign and pause lane
    control.
    self.__set_current_speed(linear_x=0.0, angular_z=0.0)
    time.sleep(3)
    self.turn_action_active = False # Clear the action sign and restore
    lane control.
    self.get_logger().info('Stop action completed')
```

4. Source Code Analysis

4.1 auto_drive.py

YOLO signpost message reception:

```
def sign_callback(self, msg:DetectionObject):
    '''Subscribe to YOLO recognition results'''
    class_name = msg.class_name
    object_distance = self._get_distance(msg.ymin) # Calculate distance
    object = Object(msg.class_name, msg.confidence, object_distance)
```

Multiple filtering mechanisms: mainly to prevent continuously recognizing road signs and triggering actions; the action will only be unlocked when different road signs are recognized.

```
# 1. Check if other actions are being performed.
if self.action_runing:
    return

# 2. Check cooldown time (to prevent repeated triggering)
current_time = time.time()
if class_name in self.last_processed_time:
    time_diff = current_time - self.last_processed_time[class_name]
    if time_diff < self.cooldown_time: # Default cooldown: 3 seconds
        return

# 3. Check if it is a duplicate road sign.
if class_name == self.last_sign:
    return

# 4. Check if the road sign is enabled.
if not self.sign_enable_list[对应的索引]:
    return
```

Event queue processing

```
def handle_events(self):
    '''Handling target recognition events'''
    while True:
        if not self.event_queue.empty():
            object = self.event_queue.get()
            self.action_runing = True
            current_sign = object.class_name

            try:
                # Perform the appropriate action based on the type of road sign.
                if object.class_name == "straight" and
self.sign_enable_list[10]:
                    self.go_straight()
                elif object.class_name == "turnright" and
self.sign_enable_list[11]:
                    self.turn('right')
                # ... Other road sign processing

            finally:
                # Update status record
                self.last_processed_time[current_sign] = time.time()
                self.last_sign = current_sign
                self.action_runing = False
```

Specific action implementation

```
def stop(self):  
    '''停止 stop'''  
    self.turn_action_active = True # Set an action sign and pause lane  
control.  
    self.__set_current_speed(linear_x=0.0, angular_z=0.0)  
    time.sleep(3)  
    self.turn_action_active = False # Clear the action sign and restore  
lane control.  
    self.get_logger().info('Stop action completed')
```